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Patients' perspectives of emergency care quality and priorities for care improvement

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Abstract

Purpose – Studies have examined strategies implemented to strengthen quality of emergency care in healthcare provider institutions in Ghana. But few studies have focused on what determines quality of emergency care from the patient's perspective. The purpose of this paper is to fill that gap by examining factors salient to gauging quality of emergency care and priority areas for care improvement.

Design/methodology/approach – Cross-sectional data were collected from patients admitted in emergency units of public hospitals in two regions: Greater Accra and Central Regions. A structured questionnaire designed with inputs from emergency medicine physicians and patients was used to collect data from 381 patients. Principal component analysis (PCA) and logistic regression models were computed to respectively determine salient measures of emergency care quality and their association with patient overall perceived quality of emergency care.

Findings – Using the PCA, four factors (social and relational care, attentive prehospitalised care, ward quality and privacy and medical supplies) were derived as salient measures of emergency care quality. All the factors derived had statistically significant association with patient overall perception of quality.

Originality/value – Emergency care quality improvement strategies that incorporate the dimensions identified can produce effective therapeutic outcomes.

Keywords Evidence-based practice, Ghana, Healthcare quality, Patient perspectives, Emergency care, Management, Quality, Quantitative research, Prehospitalised, Inhospital

Paper type Research paper

Introduction

A critical component of healthcare delivery in the hospital setting relates to handling emergency cases with receptiveness to avoid fatalities. Compared to other units in the hospital, the emergency department is often faced with the daunting task of applying medically appropriate methods of providing initial resuscitation and stabilisation to critical patients in order to save lives. Therefore, there is growing concern that the emergency department should not lack needed human and clinical resources if mortalities associated with acute medical, surgical and obstetric conditions are to be reduced. However, in some low and middle income countries (LMICs), especially in sub-Saharan Africa (SSA), emergency care is not given priority in public health policies (Anderson *et al.*, 2012; Periyanayagam *et al.*, 2012).

The underlying causes of poor attention to emergency care in SSA are multifaceted. First, governments have over the years prioritised resources towards targeted disease programmes to reduce the burden of communicable and non-communicable diseases and their impact on population health. Second, emergency medicine is a new and emerging field (International Federation for Emergency Medicine, 2012) that is often misconstrued and conceptualised as availability of emergency transport systems as opposed to onsite care with supporting inpatient specialist (Calvello *et al.*, 2013).

Limited attention and underfunding of emergency care in LMICs have posed challenges such as overcrowding in emergency departments (Sun *et al.*, 2003), inadequate emergency transport systems (Razzak and Kellermann, 2002), lack of standardised protocols (International Federation for Emergency Medicine, 2012; Wolf *et al.*, 2012), difficulties of access to essential emergency medical services (Joshipura *et al.*, 2004; Mock *et al.*, 2006) and

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long waits (Muntlin *et al.*, 2006). Clearly, these challenges undermine the provision of high-quality emergency care. Yet, quality of emergency care is an essential component of the care seeking process, because it has direct influence on clinical outcomes of acute patients. The provision of insufficient quality of emergency care potentially influences patients to personally manage their conditions at home rather than seek formal healthcare.

Achieving high quality emergency care requires putting in place critical measures to accomplish what the Institute of Medicine refers to as crossing the "quality chasm": care provided should be timely, safe, effective and patient-centred (Alessandrini *et al.*, 2011; Kohn *et al.*, 2001). This has become necessary as many evidence-based studies somewhat support a unanimous conclusion that in many LMICs emergency care is deficient in quality to meet patient satisfaction (Cooke *et al.*, 2006; Marinsek *et al.*, 2007; Muntlin *et al.*, 2006). Optimal quality of emergency care that takes into account identification of patient needs and preferences could yield substantial dividends including prevention of avoidable deaths.

In response to the shortfalls of emergency service delivery globally, and particularly in LMICs, the World Health Organisation has called on countries to develop formal integrated emergency care systems to catalyse the delivery of effective and efficient care (Mock, 2007). This has attracted overwhelming response. For example, several emergency medicine symposia have been held across Africa with the aim of identifying possible measures to deliver time-sensitive interventions and life-saving emergency care (Wolf *et al.*, 2012). Additionally, the scope of the interventions adopted by the African federation of emergency medicine in the last few years to improve upon emergency care looks promising to the continent which suffers from the highest rates of a range of road-traffic accidents, maternal deaths and communicable and non-communicable diseases burden (Calvello *et al.*, 2013).

Quality of emergency care in Ghana has generally been described as poor and woefully inadequate to meet the express health needs of the acutely ill (Norman *et al.*, 2012). Due to service delivery gaps and limited capacity to provide timely life-saving care, 81 per cent of emergency deaths relating to RTAs occur in the prehospital setting (London *et al.*, 2002), while many other deaths relating to trauma occur either in the emergency room or few hours after arrival in the health facility (Kortbeek *et al.*, 2008). Emergency situations are further worsened by poor road network that hinder efforts at promptly transporting acute patients to health delivery points. Continuity of supplies for emergency care in Ghana is severely constrained by limited financial resources (Norman *et al.*, 2012). Critical human resources for emergency care are also lacking because emergency medicine in Ghana is still rudimentary and many personnel are without formal training in that field (Rominski *et al.*, 2011).

Emergency medicine in Ghana has only recently been given a new face lift following the introduction of the emergency medicine residency training programme by the Ghana College of Physicians and Surgeons (GCPS). It was anticipated that an estimated 50 physicians would have received specialist training in emergency medicine by 2016 (Osei-Ampofo *et al.*, 2013). In recognition of the critical shortage of emergency personnel, the Kwame Nkrumah University of Science and Technology has also introduced a bachelor's programme in emergency nursing to complement the training offered by the GCPS. These interventions are aimed at building the capacity of the health system to respond appropriately to emergency cases presenting in health facilities.

However, much more is needed to integrate patients' feedback systems into curricular training to improve the health delivery process. Also, more research directed at identifying the constituents of quality of emergency care from the perspective of acute patients is vital to inform policy advances in this field of medicine. Moreover, since poor quality of care mostly leads to poor outcomes, complementing evidence-based studies with improvement of institutional structures, processes and systems is necessary to engineer

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transformation in emergency healthcare delivery. Thus, this study seeks answers to two research questions:

RQ1. What factors are salient to gauging quality of emergency care from the acute patient's perspective?

RQ2. Which of these factors needs to be prioritised by virtue of their association with quality of emergency care?

Empirical answers to these questions will enable resources to be channelled appropriately for quality improvement in emergency units.

Conceptualising quality of emergency care

Scholars contributing to quality of emergency care have somewhat failed to converge on its component measures. Different interpretations are given to the discipline, perhaps, because indicators of quality of emergency care are context specific (Cameron *et al.*, 2011). The nonexistence of universal measures of high-priority quality of emergency care tends to prevent cross regional and country comparisons and benchmarking. Despite the variations, there is some consensus that quality of emergency care should reflect the broad spectrum of prehospital and inhospital care (Kobusingye *et al.*, 2005; Razzak and Kellermann, 2002).

Prehospitalised care is an important element of emergency care quality. Health providers must have the capacity to provide responsive and timely care that is adequate enough to resuscitate the patient. Relevant literature has demonstrated that adequate prehospitalised care that take into account diagnosis and treatment at reasonable time is crucial for patient's resuscitation and stabilisation, especially those with life-threatening conditions (Osei-Ampofo *et al.*, 2013). Responsive prehospitalised care is particularly important to arrest danger signs and complications of critical patients. Substandard quality of prehospitalised care, more especially ill preparedness on the part of providers to deliver timely responsive care is often a major cause of fatalities among critical patients in some LMICs settings (Norman *et al.*, 2012; Tiska *et al.*, 2004). Strengthening both technical and service quality at the point of arrival could contribute greatly to minimising preventable mortalities. Prehospitalised care provides an important survival link to definitive hospitalised care for the acutely ill.

Quality of inhospital care involves all treatment rendered to the patient during admission in the health facility. It is determined by both human and structural factors. The human attributes include the presence of a range of personnel mix and the interpersonal aspect of care (Dogba and Fournier, 2009). Medical personnel mix with sufficient training and skill in emergency care are able to save life-threatening conditions, prioritise care, carry out resuscitation and provide emotional support to the patient and his or her caregivers (Brysiewicz, 2011). Another important human attribute that cannot be overlooked when assessing quality of emergency care is the provider's ability to provide personalised care and engage in more meaningful relations with the patient (Grossmann-Kendall *et al.*, 2001). Health providers should strive to uphold and respect the patient's dignity (Cameron *et al.*, 2011). The study by Hsu *et al.* (2006) revealed that patients quality perception is influenced by dignity, attention and privacy in the care seeking process. Generally quality of care is enhanced when providers identify themselves more closely with patients in order to understand their needs and preferences (Oluwadiya *et al.*, 2010).

Structural factors describe the environment or setting in which care is administered (El Sayed, 2012). Variables in this context include but not limited to equipment, facilities, space, medications, supplies and specialized equipment (Razzak and Kellermann, 2002). Evidence show that the capacity of the health facility to have sufficient quantities of

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clinical resources including both drug and non-drug supplies as well as appropriate medical equipment demonstrate the potential to provide sufficient quality care (Razzak and Kellermann, 2002). Structural quality has a direct link to patient health outcome (in terms of increases or decreases in disabilities and mortalities). It also significantly affects satisfaction and the decision to return to the same facility in the future (Boudreaux *et al.*, 2004).

Methodology

Study setting

The study was conducted in public hospitals located in two coastal regions: Greater Accra and Central. Each of the hospitals had an emergency unit operating 24 hours a week for emergency cases from within and across the regions. At the time of conducting the study, some of the hospitals were without critical health human resource mix. Non-emergency medicine practitioners, nurses and physicians attended to patients in three of the hospitals. In the other two hospitals, trained physicians in the field of emergency medicine delivered care with the support of emergency nurses.

Design and data collection

The study was based on a cross-sectional design involving three months of data collection. Data were collected using a structured questionnaire. The questionnaire was designed taking into account the theoretical literature, expert advice by two physicians and inputs from a pretest conducted on 35 patients in similar hospitals. During the pretest, patients were asked to respond to the questionnaire and comment on its suitability. Consequently, some questions were dropped while others were reframed to reflect patients actual experience of quality of prehospitalised and inhospital care. The final questionnaire comprised of three sections. Patients' demographic characteristics (age, sex, marital status, marital status, employment status and condition) as well as number of days on admission were captured in the first section.

The second section of the questionnaire captured questions pertaining to the measures of emergency care quality. The questions were categorised into: nature and quality of treatment before hospitalitation (seven items); nature and quality of health providers attention to patients (six items); medical infrastructure including supplies and equipment (seven items); quality of physical features of the ward (five items); and privacy in patient-provider engagements (five items). The last section captured questions on overall perceived quality of care ranging from prehospitalised to inhospital care. These questions were dummies (1 = good; 0 = poor). In this study, prehospitalised care refers to care provided immediately upon arrival at the health facility. It does not include elements of care provided to patients before arrival to the facility.

The second author together with two research assistants administered the questionnaire. The research assistants were given prior training on interview techniques, structure and content of the questionnaire and how to approach respondents. Each selected patient was made to understand the study's purpose and outcome of interest before seeking his/her consent to participate and be interviewed. Inclusion criteria were if patient was rushed into the hospital and subsequently admitted to the emergency unit. The questionnaire was administered in English. However, for those who could not read or write the English language, questions were translated into their local language. Patients who were mentally incapacitated, unconscious, and could not recollect events that occurred on arrival to the facility were excluded. The research team visited the hospitals on working days to administer the questionnaire. At least ten working days were spent on data collection in each hospital.

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Analysis

The SPSS version 21 was used to analyse the data. The analysis was done by the first author and reviewed by a biostatistician. Descriptive statistics were used to present demographic characteristics of respondents and the number of admission days. To answer the first research question, we employed principal component analysis (PCA) with orthogonal (varimax) rotation. This statistical technique helped reduce the many number of items of the questionnaire and classified them into discrete fewer factors measuring quality of emergency care (Field, 2009; Pallant, 2010). The factor structure of the PCA was strong because of a higher Kaiser-Meyer-Olkin value of 0.818. Similarly, the Bartlett's Sphericity test was statistically significant ($p < 0.01$).

To be sure that the set of items were consistent in measuring each factor produced, we estimated Cronbach's α values. Field (2009) stated a threshold α value of 0.75 as desirable. Accordingly, all our estimations were in fulfilment of that. To answer the second research question, we turned to a multivariate logistic regression model. The model was fitted to determine statistically significant association between each of the factors of emergency care quality produced by the PCA and the dependent variable: overall perceived quality of emergency care. We controlled for patient's characteristics to eliminate confounding relationships. For each of the factors, all the items were computed into one composite measure of that factor and used for the regression estimation.

Results

Results in Table I show that the majority of the respondents were males (53.8 per cent), married (56.2 per cent) and employed in the public sector (36.1 per cent). The mean age of the patients was 36 years and average admission days was 6. A large proportion of the respondents (17.2 per cent) presented with obstetrical-related conditions, 15 per cent with RTAs, 11.3 per cent with diarrhoea related and the least number (8.4 per cent) with bronchial asthma (Figure 1).

Salient measures of emergency care quality are presented in Table II. A total of 20 items were produced and clustered around the four factors of quality. The first factor, social and relational care accounted for 44.3 per cent of the variance. This factor indicates that patients put so much emphasis on social care, relational information sharing, concern for patient condition, respect and reciprocal interactions. The second factor was attentive prehospital care which accounted for 9 per cent of the variance. This factor highlights the importance that patients attach to urgent treatment on arrival, maximum provider attention, appropriate diagnosis and assurance. The third factor with a variance of 7 per cent was privacy and ward quality. For this factor, patients were concerned about ventilation of the ward, physical environment of the ward and privacy issues. The final factor termed medical supplies produced a variance of 4 per cent. Patients' perspectives were that for quality of emergency care to prevail in the hospital, essential drugs, equipment and supplies must be adequately available.

The regression model is presented in Table III. All the four factors were statistically and significantly associated with overall perceived quality of emergency care after controlling for patients' characteristics. To be precise the overall perceived quality of emergency care had positive statistical significant association with social and relational care ($\beta = 0.13$; $p < 0.01$), attentive prehospital care ($\beta = 0.19$; $p < 0.05$), ward quality and privacy ($\beta = 0.16$; $p < 0.05$) and medical supplies ($\beta = 0.26$; $p < 0.01$). These indicate that overall perceived quality of emergency care increases with increasing effect of all four factors. Thus, all four factors must be prioritised in quality improvement strategies and processes of emergency care.

Characteristic	$n = 379$ $n(\%)$
Age	
18-24	55 (14.5)
25-34	141 (37.2)
35-44	84 (22.2)
> 44	99 (26.1)
Mean (SD)	36 (12.59)
Gender	
Male	204 (53.8)
Female	175 (46.2)
Marital status	
Married	213 (56.2)
Not married	166 (43.8)
Employment	
Unemployed	91 (24.0)
Private sector	82 (21.6)
Public sector	137 (36.1)
Self-employed	69 (18.2)
Admission days	
1-5	215 (56.7)
6-10	109 (28.8)
11-15	32 (8.4)
> 15	23 (6.1)
Mean (SD)	6 (6.417)
Mode of arrival	
Ambulance	89 (23.5)
Non-ambulance	290 (76.5)

Table I. Socio-demographic characteristics

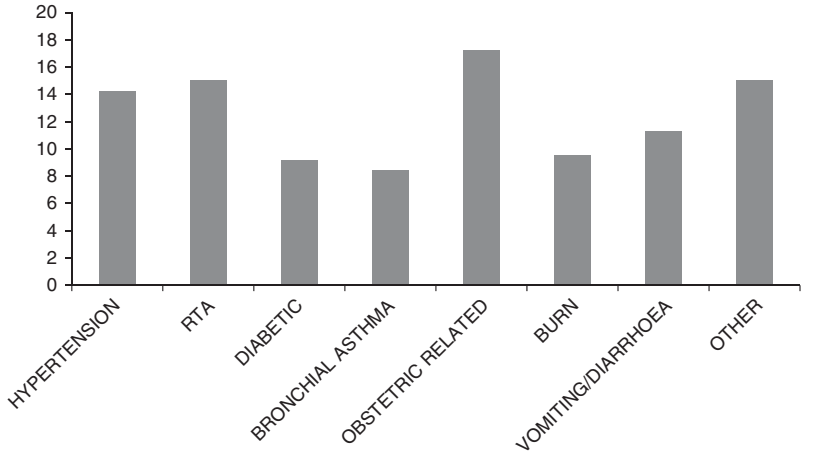

Figure 1. Conditions of patients

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Quality dimension	Factor loading	Eigenvalues	% of variance	Cronbach's α
Social and relational care		10.306	44.310	0.89
Health providers…				
Very much support and cares for me	0.614			
Interact and share vital information with me	0.564			
Show concern that I will get better	0.610			
At all times empathise with me	0.783			
Show respect for me	0.725			
Attentive prehospitalised care		1.977	8.597	0.88
Upon arrival in the hospital, I received…				
Urgent treatment	0.781			
Reasonable medical attention	0.868			
Care that was resuscitating	0.789			
Appropriate diagnosis	0.730			
Assurance that I will be fine	0.634			
Ward quality and privacy		1.511	6.569	0.91
The ward has a neat environment	0.671			
The ward is physically appealing	0.732			
Health providers give me private attention	0.842			
Health providers insist on privacy when attending to me	0.823			
Medical supplies		1.014	4.410	0.84
Essential drugs for my condition are available	0.633			
Basic supplies are available	0.654			
The hospital is resourced with appropriate equipment	0.581			
Medical equipment in the hospital are modern	0.567			

Notes: Extraction procedure: principal component analysis; rotation procedure: varimax with Kaiser normalisation; Kaiser-Meyer-Olkin measure of sampling adequacy: 0.959; $p < 0.001$

Table II. Principal component analysis of factors of emergency care quality

Covariate	Coefficient (CI: 95% CI)
Social and relational care	0.13 (0.97-1.10)*
Attentive prehospitalised care	0.19 (0.85-0.98)**
Ward quality and privacy	0.16 (0.79-0.91)**
Medical supplies	0.26 (0.95-1.19)*
Age	0.11 (0.89-1.39)
Gender	0.49 (1.03-2.60)*
Marital status	0.51 (1.0-2.62)*
Employment	0.07 (0.86-1.35)
Admission days	0.40 (1.13-1.98)**
Condition	0.07 (0.85-1.03)
Insurance status	0.38 (0.88-2.43)
Pearson χ^2(df)	46.10 (7)
−2likelihood ratio test	411.66
Pseudo R^2 (Cox and Snell)	0.211
Pseudo R^2 (Nagelkerke)	0.281

Notes: *,**Significant at 5 and 1 per cent, respectively

Table III. Regression results of factors associated with perceived quality of emergency care

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Discussion

The study's findings reflect those linking quality of emergency care to human and structural factors (El Sayed, 2012). The health provider dimension appeared in this study as a significant indicator measure of quality. Patients in this study were concerned with the social and relational aspects of emergency care. They wanted health providers to be supportive, respectful and empathetic in the care-giving process (Atinga *et al.*, 2015). The findings also suggest that quality of emergency care is not only linked to having the right health professional skill mix, but also their ability to interact more with patients in the care process (Grossmann-Kendall *et al.*, 2001). Patient-provider interaction is important as it enable cross sharing of ideas and clinical decisions, and fosters better therapeutic outcomes (Dibbelt *et al.*, 2009). Thus rather than focusing primarily on the patient's condition management, health providers must seek better ways of giving patients the opportunity to share their concerns, ask questions and be part of critical clinical decision making (Atinga *et al.*, 2016). This would require having understanding of patients' cultural orientation so that social and language barriers can be bridged. Respect for the patient emerged as a critical factor of social and relational care. Patients need to be dignified and esteemed if clinical outcomes are to have strong and positive therapeutic effect (Chadwick, 2012). Hence, health providers, including doctors and nurses, need to rethink new attitudes and practice that embrace the patient, and recognising him or her as an important partner in the clinical setting.

Prehospitalised care was found to be an important component of emergency care quality. Results of the factor structure revealed that patients prioritised care that is urgent, resuscitating and appropriate upon arrival in the facility. This finding does not come as a surprise. In Ghana, just like other LMIC countries, emergency care in the prehospital setting is sub-optimal (Banerjea and Carter, 2006; Molyneux, 2009). Apart from deficiencies relating to medical infrastructure, equipment, medical logistics and supplies (Hsia *et al.*, 2012), the human element of prehospital care often falls short of standards. Swift attention and appropriate care for patients in most health facilities are not good enough, as is health professionals capacity for triaging (Kobusingye *et al.*, 2005). As a result, there is usually a marked contrast between patients' expectations of care and what is provided in practice upon arrival. In other words, care provided does not always correspond with patients' expectations.

It was found that privacy and quality of the ward setting are vital measures of emergency care quality. Key elements of ward quality prioritised by the patients are hygienic, appealing and serene environment. This finding corroborates earlier studies. For example, in the study involving public hospitals inpatients by Atinga *et al.* (2011), it was found that the physical environmental aspect of quality was a significant predictor measure of patient assessment of quality. In that study, the authors noted that patients were critical about ward environment akin to their homes (Atinga *et al.*, 2011). Both the present and previous findings are understandable. This is because, public wards in Ghana's public hospitals are often characterised by overcrowding, poor ambience and stained walls arising out of poor maintenance culture.

Privacy appeared in this study as a significant measure of quality, as is the case elsewhere (Atinga and Baku, 2013). Patients preferred care in which health providers maintained privacy during consultations and discussions. In a clinical environment like emergency, privacy is more important because patients suffer from varying degrees of conditions, and might require a private environment to divulge vital information that can aid diagnosis and treatment. For this reason, patients may become reticent to share vital information if there is no maximum privacy (Atinga *et al.*, 2015).

Finally, supplies and essential medical equipment for care delivery is an important quality factor to the patient. This finding suggests that essential drugs available and in sufficient quantities demonstrate the capacity to respond efficiently to patient care needs.

In the context of emergency care, the significance of having in stock sufficient drug supplies has been emphasised (Norman *et al.*, 2012). It increases health provider responsiveness and attention to patients, and enhances patient experience and satisfaction, all of which facilitate therapeutic outcomes. Poor stock level increases risk of complications and mortalities (Razzak and Kellermann, 2002). On the aspect of medical infrastructure, it is apparent patients are concerned that emergency departments need to have relevant equipment that are also modern. We find this important for a number of reasons. First, availability of essential equipment signals hospital preparedness to handle acute and complicated case load without disturbances to routine care delivery (Norman *et al.*, 2012). Second, without appropriate modern equipment, quality of diagnosis and treatment could be undermined, leading to prolonged stay and poor therapeutic outcomes.

Conclusion and practice implications

The significance of providing quality emergency care has taken on added urgency in recent years with the impetus of policy initiatives across SSA. Within the Ghanaian context, policy tools have prioritised strengthening institutional training programmes and improvement of the technical aspect of quality (Carter *et al.*, 2012; Tiska *et al.*, 2004). Some health facilities are also assisted to implement best practice functional quality of care drawn primarily from the prevailing wisdom of western countries. Although a novel initiative, the importation of western elements can be remote from local contextual underpinnings. Second, they might be less accurate than implementing hospital-based quality improvement initiatives drawn from patients.

Motivated by the paucity of research on what counts as quality from acute patients' perspective, we sought to examine how quality of emergency can be measured, and identify priority areas for quality improvement. We were able to adduce four factors salient to patient perspectives of quality of emergency care. We also demonstrated that the four factors (social and relational care, attentive prehospital care, ward quality and privacy, and medical supplies) produced were statistically and significantly associated with patient overall perceived quality of emergency care. Based on the findings, we propose a range of options for improving quality of emergency care.

The findings suggest the need for training and skill building of health providers to enable them improve upon the quality of social and relational elements of care. More specifically, elements of patient-centredness should be embedded in educational and didactic programmes, and emphasis placed on respect, empathy and two-way communication with patients, as key to building strong bond between patients and providers in the care process. Establishing linkages of relational care with therapeutic outcomes should constantly be highlighted. Moreover, all health providers, especially emergency medicine practitioners, need to understand that quality is an art that takes into account changes in habits, personal values, attitudes and practices toward the patient.

It was found that patients regarded prehospitalised care as an important quality variable. This has several practice implications. It highlights the need for health providers to deliver prehospitalised care with minimal waiting time. In addition, capacity building of health providers is important to enable the provision of prompt care, that is, resuscitating, stabilising and life-saving. On the issue of ward quality, privacy and equipment, we suggest investment in routine maintenance to enhance face value of wards and physical surroundings, and timely replacement of old equipment. This should form part of planning and procurement processes. As noted by Hsia *et al.* (2012), it is important to have in place, systems to support functioning, maintenance and replacement of equipment. Funds and personnel with technical knowledge in the handling and repair of biomedical machines are required. Finally, maximum privacy and confidentiality should be present in the interface between providers and patients in the care-giving process.

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